Phase 4.25: Rank-32 Congruence Admission Investigation

Tactix development report for Andrew

27 August 2026

1 Outcome

The isolated proposal to remove Tactix's Congruence new-material admission gate is rejected and reverted. It is locally attractive on immutable Rank 32: the three-wave median falls from 410.760 to 405.042 seconds. It is not safe to promote across the declared benchmark population: the complete B28+C5 gate has no correctness failure or solve loss, but its PAR-2 ratio is 1.208552, above the frozen 1.10 limit. On the dominant witness, `mod4block`, runtime increases from 19.846 to 237.094 seconds and conflicts increase from 552,580 to 5,488,183.

Source-backed tracing localizes the first candidate-owned consequence. At an exact common state near conflict 105525, the candidate admits one additional Congruence round. Its 152 candidates are in a complete recorded order, all are classified `no-antecedent`, and the exact normalized cost is 456 ticks. It learns nothing and leaves the normalized logical state unchanged. The mandatory dense-to-sparse binary-watch replay nevertheless changes physical propagation order before Backbone. The first conflict-clause difference follows at conflict 105527 and the first durable learned-clause difference at conflict 105568.

This is not an implementation bug that can be repaired by suppressing the pose change. Current Kissat performs the same dense resume in Congruence, and Tactix's replay routine deliberately implements Kissat's saved-pair order. Deferring or skipping it would merely restore the faster non-reference baseline behavior. The change can be reconsidered after an earlier common- state convergence removes this sensitivity, but it cannot be accepted now. No fuzz campaign was run because the candidate failed the pre-fuzz gate and was not promoted.

2 Immutable evidence

The accepted Tactix predecessor remains source commit `3c3164dcd35b4286d3fe1aea61e4731cc8545fe3`, Release binary SHA-256 `f2eecac1...dcf`. Current instrumented Kissat remains source commit `8af8e56f174b778aef3aa45af9f739b2a5f492c2`, binary SHA-256 `b6a9b1d0...44d5`. Rank 32 uses immutable input SHA-256 `038955b1...e7f3`. The rejected solver diff SHA-256 is `7e755ac5...9953c`; its Release binary is `3f88c82d...d59b`.

Read-only Rank-32 campaigns `20260827-0710`, `0720`, and `0731` first established that the accepted Tactix and current Kissat remain exact through the early search prefix, then identified both Tactix Congruence offers in the first consequential Probe phase as declined solely by the local new-material proxy. Candidate prefix campaign `0740` proves the added round itself matches Kissat exactly for 542,604 ordered candidates, 1,627,812 normalized ticks, candidate dispositions, resulting counts, first post-phase decisions, and learned records through conflict 3073.

3 Rank-32 timing

Frozen campaign `benchmarks/runs/20260827-0750-tierd-rank32-congruence-admission-timing` uses nine independent tasks on node018, one solver invocation per task, at `-c 12 --mem=48000M`, without

RLIMIT_AS. All nine DRAT proofs verify independently, and cleanup/certificate residue is zero.

Arm	Wave 1	Wave 2	Wave 3	Median (s)
Candidate	405.042	402.648	406.604	405.042
Accepted baseline	410.619	412.313	410.760	410.760
Current Kissat	290.319	290.619	290.384	290.384

Candidate/baseline paired ratios range from 0.976559 to 0.989882; their median ratio is 0.986418. Candidate/Kissat median paired ratio is 1.395162. Within-arm spreads are at most 0.9767%, so the roughly 1.4% Rank-32 gain is stable, but it remains above the desired 400-second marker. Summary SHA-256 is a855c1c1...584.

4 Complete gate rejection

Frozen campaign benchmarks/runs/20260827-0760-BC-rank32-congruence-admission-guard uses the complete B28+C5 set under the 1200-second, 16-GiB contract. The corrected array/aggregate jobs are 1652137/1652138. The earlier dependency- pending jobs 1652135/1652136 were canceled before any solver task started because their copied aggregate lacked the declared PAR-2 assertion.

All 66 corrected tasks are valid: 34 SAT models and 32 UNSAT proofs are independently checked. There is no solve loss, invalidity, cleanup failure, or certificate residue. Correctness is therefore PASS. Performance is FAIL:

Measure	Accepted baseline	Candidate	Candidate/baseline
PAR-2 seconds	1,384.008	1,672.646	1.208552
mod4block seconds	19.846	237.094	11.947
mp1-klieber seconds	56.356	109.187	1.938

The candidate arms happened to run on node018 and predecessor arms on node017, but the mod4block candidate performs 5,488,183 conflicts and 262.9 million propagations versus 552,580 and 23.4 million. This deterministic work increase rules out placement noise as the cause. Gate summary SHA-256 is e948163d...012.

5 Causal localization

Campaign 20260827-0770 proves candidate and accepted baseline traces are byte-identical through conflict 100,000: 94,555 conflict records, 91,685 learned records, and three admitted Congruence rounds. The current-Kissat comparison has a pre-existing Tactix/Kissat learned-state split at conflict 2515, common to both Tactix pins and unrelated to this proposal.

The wider observers 0771 and 0773 saturated their event limits before the target and are frozen with that limitation. Checkpoint observer 0772 localizes the first candidate-only round to the next Probe: at the first completed boundary both arms are at conflict 120,512, but candidate has four Congruence rounds, 427,487 decisions, and 4,042,238 propagations versus baseline's three rounds, 422,586 decisions, and 4,145,226 propagations. Both have zero Congruence merge and unit yield. One terminal checkpoint task solved SAT before its nominal observer stop and returned valid solver exit 10 while the wrapper expected UNKNOWN exit 0; this is explicitly classified as a harness-contract mismatch, not a solver failure.

Narrow campaigns 0774 and 0775 capture the exact boundary. At conflict 105525 the candidate executes congruence-dense-resume; the baseline skips it and enters Substitute/Backbone directly. The

full candidate stream has 152 ordered candidates, all `no-antecedent`, with exactly 456 ticks and no logical output. Both trajectories remain exact through conflict 105526. The first conflict record differs at 105527; intermittent resyncs follow, but the first durable learned mismatch is at conflict 105568.

Kissat source `congruence.c` calls `kissat_resume_sparse_mode` after extraction, before Substitute and Backbone. Tactix source `kernel_canonicalize_binary_watches` documents and tests the corresponding global pair replay. Baseline instead delays the same deduplication until Vivify's dense entry, after Backbone. Thus the candidate's physical consequence is reference-authored. The severe regression comes from applying that reference schedule to a Tactix state that already has an earlier logical difference; it is not a collection micro-optimization opportunity at this boundary.

6 Decision and next action

The production source, tests, scheduling checker, and parity ledger have been restored to the accepted predecessor. The negative evidence remains in the frozen manifests and living progress authority. No candidate code remains, so no consolidation fuzz or correctness repair workflow is applicable.

Commit and push this evidence-only closure with a clean synchronized tree. Then rerun the immutable Rank 1–40 screen using accepted Tactix and unchanged current Kissat: 80 independent solver tasks, one invocation per task, `-c 12 --mem=48000M`, no `RLIMIT_AS`, 1200-second solver limit, and independent model/proof checking. Select the largest verified UNSAT gap from that fresh screen, then confirm it in three paired timing waves before changing behavior.